J. R. Billingsley

At time of writing —
Manager,
Cowichan Copper Co. Ltd.,
River Jordan, B.C.

# Underground Mining and Milling at Cowichan Copper Co. Ltd.

Annual General Meeting, Edmonton, April, 1963
Transactions, Volume LXVIII, 1965, pp. 87-93

**ABSTRACT**

**Of the twelve known ore zones on the Sunro property, acquired by Cowichan Copper Co., only the "B" and "C" orebodies of the "River Zone" had been explored sufficiently to indicate the grade and tonnage.**

**The "B" zone was selected for initial production. The portion of this orebody above the main adit level was developed for horizontal-ring, longhole mining. Ore from the stope would be scraped directly to the coarse ore bin.**

**The lack of a suitable mill site on surface, as well as the 8,000-ft. tramming distance, led naturally to the installation of the entire crushing and milling plant underground.**

## Introduction

IN November, 1960, a lease was obtained by Cowichan Copper Co. Ltd. (N.P.L.) from Sunro Mines Ltd., covering the main ore (River) zone of the Sunro property.

A total of twelve zones had been located on the property. Nine of these lay within the leased area. With the exception of the "B" and "C" orebodies of the River Zone, there had been insufficient exploration for tonnage and grade calculations. Exploration by previous companies had resulted in an indicated and inferred estimate of 937,000 tons of 1.80 per cent copper in the River Zone (727,000 tons in the "B" orebody and 210,000 tons in the "C" orebody). The precious metal value was approximately 35 cents per ton. A 5 per cent dilution figure was used in the tonnage and grade calculations.

Lack of a suitable mill site on surface, as well as the 8,000-foot tramming distance, led to the installation of the entire crushing and concentration plant underground.

The proposed seram drifts for the portion of "B" orebody above the adit level could feed directly to the crushing plant, and the material could then be conveyed to a suitably located mill. The ore below the level could be hoisted and conveyed a short distance to the crushing plant. No tramming would be necessary for the tonnage on which production had been predicated.

The area selected for the mill and crushing plant rooms was in solid competent basalt. The examination of drill core and visual inspection of the adit at the proposed location showed the area to be free of major faults, shears or fractures. There was no evidence of groundwater in the vicinity.

The pumping of the tailings 13,000 feet to tidewater was investigated and presented no particular problems, especially with the development of abrasive-resistant, long-life plastic pipe.

**The Mine Portal at the 5130 Elevation
Cowichan Copper Co. Ltd.**

A considerable disadvantage was the lack of stockpile space for development muck on surface. There was very little room at the dump, and the economies of returning ore underground were doubtful. Hence, the only development work completed when the first mill circuit went into operation was in the "B" stope service raise and in the stub drifts from it at scram and sub-level locations. Mill production would be limited to the development tonnages until full mine production was realized. The use of horizontal rings drilled from raises centrally located in the "B" south and north stopes on either side of the pillar raise was selected as the longhole method requiring the least amount of development and giving the quickest and most economical maximum production.

On August 1, 1961, excavation work for a two-circuit, 1,500-t.p.d. mill commenced. Concentrate was produced by the first circuit on May 1, 1962.

The second circuit was placed in operation on August 11. On June 19, the first of the horizontal longhole rings was blasted in "B" north stope. By the end of October, sufficient longholing had been completed to supply and maintain both circuits of the mill.

## Location and Topography

The Sunro mine is located on the south end of Vancouver Island, near the village of River Jordan. It is approximately 45 miles, by paved highway, west of Victoria, B.C. The portal, about 1 mile from tidewater and ¾ mile from the highway, is located on the edge of the Jordan river at the beginning of a steep 2-mile canyon.

The property is in an area of rugged and precipitous topography. The main ore zones outcrop in the canyon at an elevation of 600 feet, and are approximately 1½ miles up from the portal.

## History

The properties (Gabbro and Sunloch) were originally staked in 1915 by Mr. George Winkler, of Victoria, B.C.

In 1917, Winkler bonded the Sunloch property to the Sunloch Mining Company, which company built a narrow-gauge railroad along the east wall of the canyon to the main showings, drilled the first diamond drill holes and did the first underground work.

In 1919, The Consolidated Mining & Smelting Co. of Canada Ltd. acquired control of the Sunloch company. A total of 3,776 feet of adit was driven on the River, Cave and Centre zones, and 3,470 feet of diamond drilling was carried out.

In 1949, Hedley Mascot Gold Mines Ltd. optioned the Sunloch Mining Company property from Cominco and optioned the adjoining Gabbro property from Gabbro Mines Ltd. (controlled by Noranda Mines Ltd.). The work done by Hedley Mascot on the property consisted mainly of diamond drilling. A total of 9,354 feet was reported for 1949 and 4,082 feet for 1950.

In 1955, Cominco and Noranda Mines Ltd. consolidated the Sunloch and Gabbro companies under the name of Sunro Mines Ltd.

In 1957, Sunro Mines Ltd. collared and drove 7,841 feet of 9- by 9-foot adit on the 5100 level (100 feet above sea level). The adit roughly paralleled the river. At a point 7,500 feet from the portal and 550 feet below the canyon outcrop, the "B" orebody was intersected. A 9- by 9-foot drift was driven 360 feet along strike. From the end of this drift and from the end of the adit, 30,000 feet of diamond drilling was carried out on the "B" and "C" orebodies of the River Zone and the Cave Zone.

## Geology

The Sunro copper deposits are unique in that they are the only known orebodies of significance that occur in British Columbia in rocks younger than the Mesozoic. They occur in Tertiary (Eocene) basalts or in Tertiary (Oligocene) gabbros which intrude the basalts. Both rocks are overlain by flat-lying Tertiary (Miocene) sediments (sandstone). The orebodies appear to have a genetic relationship to the gabbros.

The ore occurs in shear or shatter zones which strike northwesterly. The Leech River overthrust outcrops about 1½ miles to the north and dips steeply northward. It separates the unmetamorphosed Tertiary rocks on the south from the pre-Tertiary metamorphic rocks on the north, the latter having been pushed up and over the Tertiary rocks. There may be a relationship between the shear zones of the orebodies and the Leech River overthrust.

The River Zone, which was the first to be mined, occurs in a shear or shatter zone in basalt, approximately 400 feet northeast of a gabbro intrusive in which the Cave Zone occurs. The River Zone is characterized by numerous steeply dipping slips or faults which converge at the northwest end. The slips are frequently slickensided and often display breccias. The more numerous the slips, the better is the ore. The best ore, up to 100 feet in width, occurs in the north end where the slips converge. As the slips diverge to the south, the ore becomes leaner and finally tails off into separate lenses, each of which lies along one of the stronger slips and may persist for several hundred feet in length. The chief ore mineral is chalcopyrite. It occurs in and along the slips, as fracture fillings near the slips or as disseminated sulphides between the slips. Native copper is common in the graphitic portions of the slips.

The zone is contained within an envelope of soft, shattered and hornblendized basalt. Outside the envelope, the basalts are fresh and massive and exceedingly strong.

## Crusher and Mill Excavation

### *Ventilation*

Prior to excavation for crushing and milling, an 8- by 8-ft. raise was driven vertically 570 feet, using an Alimak raise platform. From this, at a height of 550 feet, a 7- by 7-ft. drift was run 100 feet out to surface 60 feet above the river. This raise, in addition to providing future ventilation, is used as a service raise, down which were brought the air, water and power lines.

During the driving of this raise, ventilation was provided by a 440-volt, 24-inch Woods single-stage fan, delivering 24,000 cubic feet per minute from the portal through 24-inch polythene tubing strung on messenger wire 7,600 feet along the adit.

Compressed air was supplied by two 600-c.f.m. portable compressors, located near the portal.

### *Excavation*

(See Figure 1)

For the crusher room, pilot drifts were first driven along the perimeter of the area to be excavated. Slots were then cut to full dimensions at Number 1 and Number 3 crushers. Slashing was carried out following the sequence shown in Figure 2.

For the mill room (Figure 3), slots were established and the arched back slashed for the full length, paralleling and including the main adit. The remaining benches between slots were drilled off with a G.D.P.R. 123 drill, mounted on a standard G.D. air trac. The size of hole was 3 inches and the average length 60 feet. An Eimco 21B mucking machine and an Eimco 630 crawler were used for mucking into 90-cubic-foot Granby cars. A 75-hp. diesel on 24-inch track was used for haulage.

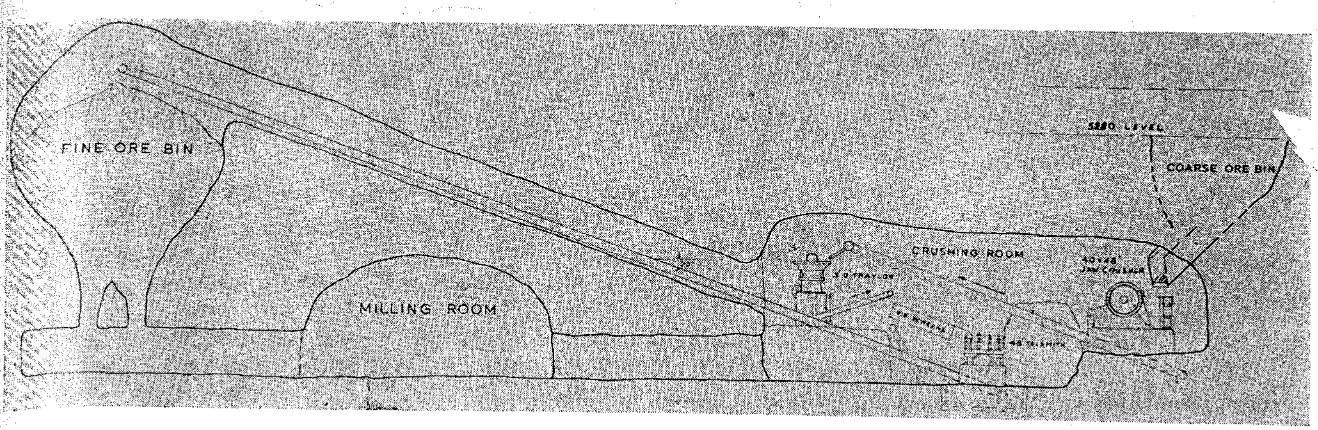



Figure 1. —Longitudinal Section.

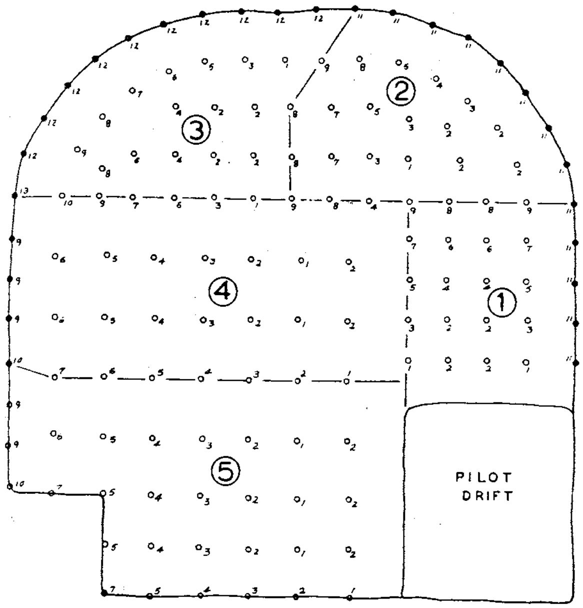



Figure 2. —The Crusher Excavation.

| S.P. | Caps | S.P. | Caps |
|---|---|---|---|
| 1 | 150 | 7 | 150 |
| 2 | 150 | 8 | 150 |
| 3 | 150 | 9 | 150 |
| 4 | 150 | 10 | 150 |
| 5 | 150 | 11 | 150 |
| 6 | 150 | 12 | 150 |

Total Rock = 9,100 tons
Powder = Cilgel-B 70% 1″ x 8″ — 6,600 lbs. or 130 cs.
Xactex — 560 lbs. or 16 cs.
Powder factor = 0.79 lb./ton

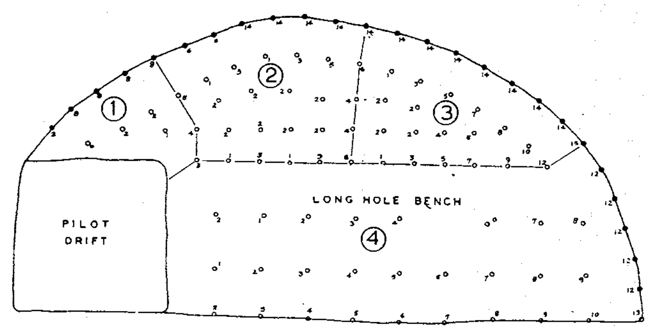



Figure 3. —The Mill Excavation.

| S.P. | Caps | S.P. | Caps |
|---|---|---|---|
| zero | 150 | 8 | 100 |
| 1 | 150 | 9 | 50 |
| 2 | 150 | 10 | 50 |
| 3 | 150 | 11 | 50 |
| 4 | 100 | 12 | 150 |
| 5 | 100 | 13 | 100 |
| 6 | 100 | 14 | 100 |
| 7 | 50 | 15 | 100 |

Total tonnage — 10,500
Cilgel-B — 70% 1″ x 8″ = 3,920 lbs. or 80 cs.
Xactex — 775 lbs. or 22 cs.
Amex II — 6,700 lbs.
Over-all powder factor = 1.09 lbs./ton

The arched back of the excavation was roof bolted as the top slice progressed. The bolts, 5/8 inch by 6 feet, with 6- by 6- by 1½-in. plates, were set on a 4- by 4-foot pattern.

Perimeter holes, drilled at 24-inch centers, with a 30-inch burden, were blasted with Xactex.

The egg-shaped fine ore bin, central to and just outside the mill excavation, was excavated by raising and shrinkage stoping to the desired size. The coarse ore bin, between the jaw crusher and the ore, was excavated in a similar manner.

Gunite was used in two locations — the back of the transformer room (30 by 25 feet, north part of the mill excavation) and in the cut-out for the Ross feeder above the jaw crusher.

Access and conveyor heading costs are shown in Table I, excavation costs in Table II. These costs include all direct and indirect charges for excavating, mucking, tramming, drilling, roof support, etc.

**Table I—Access and Conveyor Heading Costs**

| | Feet | Total | Cost/Ft. |
|---|---|---|---|
| Pilot Drifts (8′ x 8′) | 263 | $20,982.72 | $79.77 |
| Conveyor Raise (7′ x 7′) to Fine Ore Bin | 136 | 6,157.86 | 45.27 |
| Feeder Drifts (7′ x 7′) | 104 | 7,027.18 | 70.00 |
| **Total** | **503** | **$34,167.76** | **$67.92** |

**Table II—Excavation Costs**

| Location | Cubic Feet | Total Cost | Cost/ Cu. Ft. |
|---|---|---|---|
| Coarse Ore Bin (including raise) | 22,000 | $13,381.56 | 60.8¢ |
| Fine Ore Bin (including raise) | 27,820 | 11,073.36 | 36.2¢ |
| Crusher Room | 120,759 | 35,760.00 | 21.3¢ |
| Mill Room | 122,766 | 56,712.83 | 46.2¢ |
| Mill Shop | 6,475 | 5,819.83 | 90.0¢ |
| **Total** | **299,820** | **$122,747.58** | **40.9¢** |

## Construction

Lights and power for construction were supplied by a 30-kw. diesel electric generator, set up in the crusher chamber. Primary voltage was supplied at 440 volts to drive a cement mixer and a portable concrete conveyor.

Gravel and cement were trammed to the pouring site. The mixer was positioned so that gravel was mucked directly from the cars to the mixer hopper. The conveyor, 25 feet in length, delivered concrete to the forms. An 11-cu.-ft. mixer was used throughout. All pours were tamped with an air vibrator. A total of 619 cubic yards of concrete was placed; 516 yards in foundation and 103 yards in floor slabs.

The cost of the concrete placed, including all labour, services and supplies, both direct and indirect, from clearing for foundation to stripping of forms, was $95.40 per cubic yard for the foundation and $52.69 per cubic yard for the floor slabs. The cost per square foot of space, including excavation and floor slab, was $9.23 per square foot for the mill and $13.46 for the crushing plant.

It was necessary to dismantle some equipment to transport it through the adit. The ends were removed from the 7- by 10-foot rod mills and re-riveted to the shell in the mill room. The jaw crusher was sectional-framed and did not present any difficulties in moving or installation. The gyratory crushers were transported underground in three pieces. The 5- by 20-foot ball mills and the flotation cells were transported and installed as complete assemblies.

For the lifting of equipment, 1⅛- by 3-inch eye bolts were grouted in the back where required. The Perfo Sleeve system was used to cement them solidly into place.

## Milling

### *Crushing Plant*

(See Figure 4)

Muck from the coarse ore bin is reduced in three stages to a 85 per cent minus ¾-inch product.

A 48-inch Ross chain feeder regulates the run-of-mine feed to the primary jaw crusher, which is a 40- by 48-inch Birdsboro Buchanan flat belt, driven by a 160-hp. motor. The minus-6-inch crushed ore is conveyed, at 200 feet per minute, by a 36-inch belt to a 42-inch, type "K", Symons bar grizzly. The oversize from the vibrating grizzly is gravity fed to a 36-inch, type T. Y. Traylor gyratory crusher, V-belt-driven by a 100 hp. motor. The 2-inch product from the gyratory, along with the undersize from the grizzly, is conveyed at 200 feet per minute by a 30-inch belt to a double-deck, 4- by 12-foot Tyroc vibrating screen. (A 2-inch-square scalping screen is used, over a ¾-inch-square lower screen). The oversize from the screen is fed to a 48-inch Telesmith Gyrosphere cone crusher, V-belt-driven by a 125 hp. motor. The undersize, along with the product from the Gyrosphere, is conveyed at 250 feet per minute, on a 30-inch belt, to the 2,500-ton-capacity fine ore bin.

The ore is very hard and abrasive. Wear on the manganese jaw plates is 0.023 lb. per ton crushed. Concaves and mantles on the Telesmith crusher average 0.0165 lb. per ton.

Tramp steel is removed by magnets. One is mounted on Number 1 conveyor belt from the jaw crusher; another on Number 2 conveyor belt from the gyratory crusher.

A Number 19 Sheldon XS industrial exhaust fan, of 8,000 c.f.m. capacity, in series with three Sheldon 6B high-efficiency cyclones, is used for dust collection. Dust from the cyclones is returned to the Num-

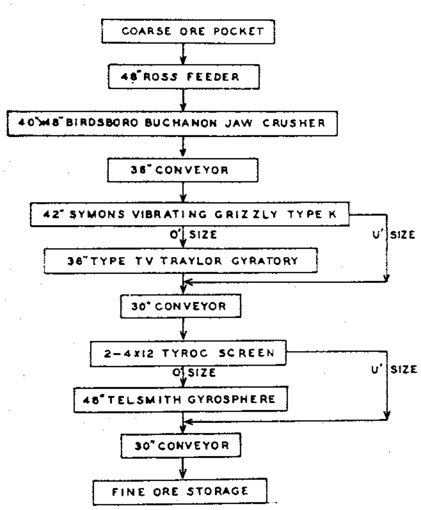



Figure 4.—(above)—The Crushing Plant Flowsheet.

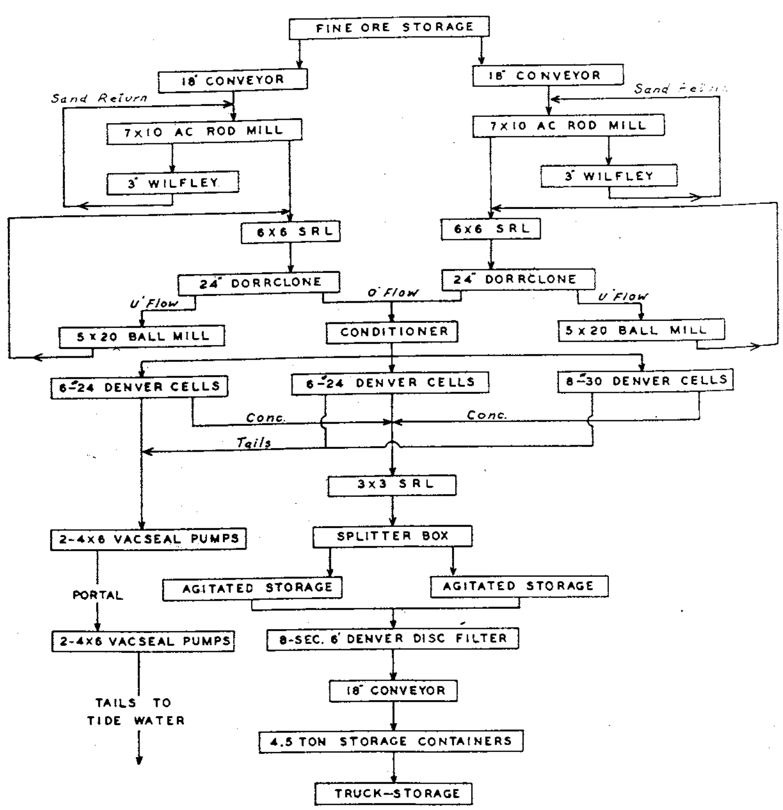



Figure 5.—(right)—The Flotation Plant Flowsheet at Cowichan Copper Co.

ber 3 conveyor belt leading from the cone crusher to the fine ore bin. Fan discharge is conducted up Number 1 service raise and discharged on surface.

### *Grinding*

*Primary*

Feed for the two identical grinding circuits is drawn from the fine ore bins by four chutes feeding onto two 18-inch variable-speed conveyor belts. Sands from the rod mill discharge are returned and fed with the crushed ore into a drum feeder. The sands, amounting to 5 to 10 per cent of the new feed, are necessary to induce maximum tonnage into the two rod mills. The primary grind mills are 7- by 10-ft. ball mills, converted to rod mills. The mills run at 25 r.p.m., or 80 per cent of critical speed, and are V-belt-driven by 250-hp., 2-300-volt motors. Rods 3 inches in diameter were used initially, but 3½-inch rods are now being tried to reduce the amount of reject material. A 45 per cent volume charge is maintained by a weekly addition of 1¼ lbs. of rod per ton ground.

In each circuit, the rod mill discharge is laundered to a sump, common with secondary grind discharge. Two 6 by 6 S.R.L. pumps (one spare), driven by 30-hp. motors and equipped with variable-speed sheaves, pump the discharge to a 24-inch Dorrclone that is in closed circuit with the two secondary mills. The Dorrclone feed is at 65 per cent solids and a pressure of 7 to 9 p.s.i. The overflow is from 38 to 42 per cent solids; the return is 80 per cent solids.

*Secondary*

Two 5- by 20-foot Worthington ball mills are used for secondary grinding on each circuit. The mills run at 24.5 r.p.m., or 80 per cent of critical speed, and are V-belt-driven by 150-hp., 2,300-volt motors. Ball charge is maintained by the daily addition of 300 lbs. of 1½ forged steel balls per mill. The secondary grind discharge averages 72 to 76 per cent minus 200 mesh.

Ni-Hard wave liners are used in the ball mills, manganese liners in the rod mills. Recent replacement of the rod mill shell liners showed wear to be 0.205 lb. per ton ground.

The screen analyses are shown in Table III.

### *Flotation*

(See Figure 5)

The Dorrclone overflow (38 to 42 per cent solids) is gravity fed to an 8- by 8-foot conditioner tank where the majority of the reagents are added. The pulp is metered and gravity fed to three banks of Denver cells in parallel. There are two banks of six Number 24 cells and one bank of eight Number 30 cells. Over 20 minutes of flotation time is provided at 1,500 tons. The two Number 24 banks are set up for single-stage cleaning, the Number 30 cells for two-stage cleaning.

The reagents used comprise 1.4 lbs. of lime, to a pH of 10.6, 0.10 lb. Z6 or CF 51, 0.04 lb. Dowfroth 250 or CF 81, and 0.02 lb. of NaCN. With feed averaging 2 per cent copper, a recovery of 96.08 per cent is achieved. The concentrates average 25 to 26 per cent copper.

### *Filtering*

Concentrates of 35 to 40 per cent solids are pumped by a 3 by 3 S.R.L. pump to two 8- by 10-foot agitated storage tanks. A Dorrco Triplex diaphragm pump raises the concentrate to an eight-disc, 6-foot Denver filter. A 24-inch vacuum is maintained by an 8- by 24-inch I-R vacuum pump. The filter cake has a moisture content of 10.20 per cent.

### *Concentrate Handling*

Cake from the filter drops to an 18-in. reversible belt conveyor and is conveyed into 3- by 4- by 6-ft. high-tensile-steel containers on 24-in.-gauge flat cars. These conveyors are loaded directly (Figure 6) from either end of the conveyor. The containers hold 4.1 tons of concentrate each and are normally trammed five at a time. They are stored in a covered open-end storage shed. An electric hoist on a monorail is

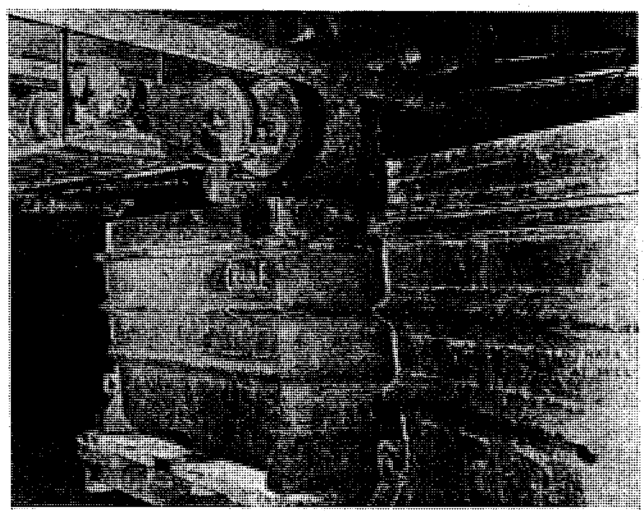


**Figure 6. —Loading Concentrate at the Mill.**

**Table III—Screen Analyses**

| Material | +¾ | +½ | +¼ | +7 | +10 | +20 | +30 | +60 | +80 | −80 | +100 | +150 | +200 | −200 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Feed to Rod Mill | 16.60 | 24.29 | 37.41 | 7.00 | 3.32 | 4.05 | 1.69 | 1.44 | 0.56 | | 0.42 | 0.40 | 0.45 | 2.36 |
| Rod Mill Discharge | | | | 1.20 | 3.35 | 23.00 | 18.00 | 17.05 | 5.10 | 32.10 | | | | |
| Ball Mill Discharge | | | | | | | | 6.50 | 7.60 | | 8.85 | 10.90 | 13.25 | 52.90 |
| Dorrclone O'Flow | | | | | | | | 0.50 | 2.10 | | 3.70 | 6.70 | 10.30 | 76.70 |
| Tailings | | | | | | | | 0.15 | 0.80 | | 2.65 | 5.85 | 11.20 | 79.35 |
| Concentrate | | | | | | | | 0.30 | 1.05 | | 1.40 | 2.20 | 5.70 | 87.35 |

used to remove the containers from the mine cars and load them on 65-ft. truck trailers. They are lifted by a cradle that engages two offset studs located near the bottom of each container. For loading, the cradle is locked to prevent the container from tripping. To unload, the container is hoisted into position and tripped. Five containers per trip are hauled 58 miles to the wharf site.

### *Tailings Disposal*

Tailings are pumped 13,000 feet from the mill to tidewater. Two 4- by 6-ft. Vacseal pumps in series, driven by 50-hp. motors, pump from the mill 8,000 feet to the portal, where an identical pumping arrangement forces the tailings over a 150-foot vertical height and along a 5,000-foot horizontal distance to disposal. A velocity of 6.5 feet per second is maintained by adjustment of make-up water in the tailings sump in the mill. The tailings are handled at 24 per cent solids. The outlet pressure at the pumps is 105 p.s.i.

Six-inch plastic (P.V.C.) pipe is used throughout. The 8,000-foot portion in the adit is on a favourable grade of 0.25 per cent. The pressure at the portal discharge sump is approximately 40 p.s.i. Automatic flushing equipment is maintained at both pumping stations to prevent line blockage in case of power failure.

Very little maintenance or repair has been required on the tailings line and pumps. The cost to date for maintenance, power and operation is 5 cents per ton milled.

## Longhole Mining

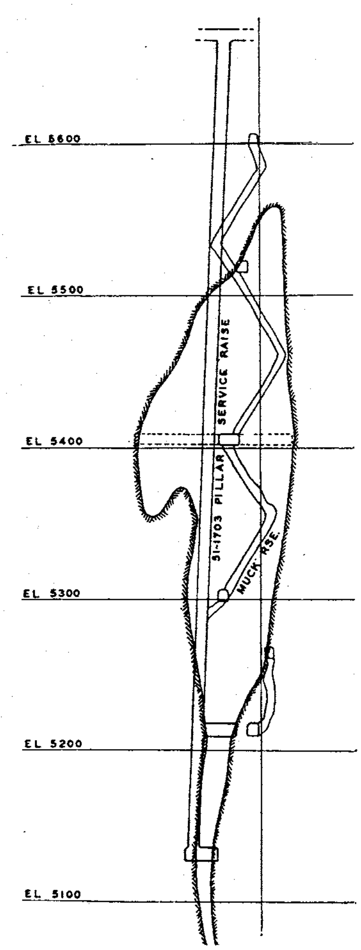
### *Stope Preparation*

The "B" orebody above the scram level was divided into two stopes, north and south, by a 50-foot pillar (Figure 7). Ventilation, service and entry are effected through a 550-foot, 8- by 8-foot raise through the pillar. The Alimak raise platform was used for the driving and timbering of the raise. A 5- by 5-foot raise for development muck was driven conventionally from sub-level to sub-level. This raise, chute controlled, feeds into the coarse ore bin. Sub-levels were established at 100-foot vertical intervals.

Scram drifts, 8 by 8 feet in section, were driven on the 5210 level. Draw points were established at 25-foot centers, starting 25 feet from the pillar. Raises were taken up 25 feet in the draw points and coned to break through to adjacent draw points.

*North Stope*

In the north stope, a 12-foot-wide slot was established from wall to wall at the first draw point. Then, an 8- by 8-foot drift was driven along the west contact of the stope 41 feet above the scram. The remainder of the undercut (see Figure 8) was longholed from the drift. Jack-leg drills were used to trim up irregularities in the east wall following the longhole blasts. The longhole rings were at 5.5-foot centers, with 8-foot toe spacing. The 2⅛-in. holes were drilled with a G.D.C.F. 99 machine, averaging 100 feet per man-shift. The tonnage broken was 2.32 tons per foot drilled, with a powder factor of 0.34 lb. of Amex II.



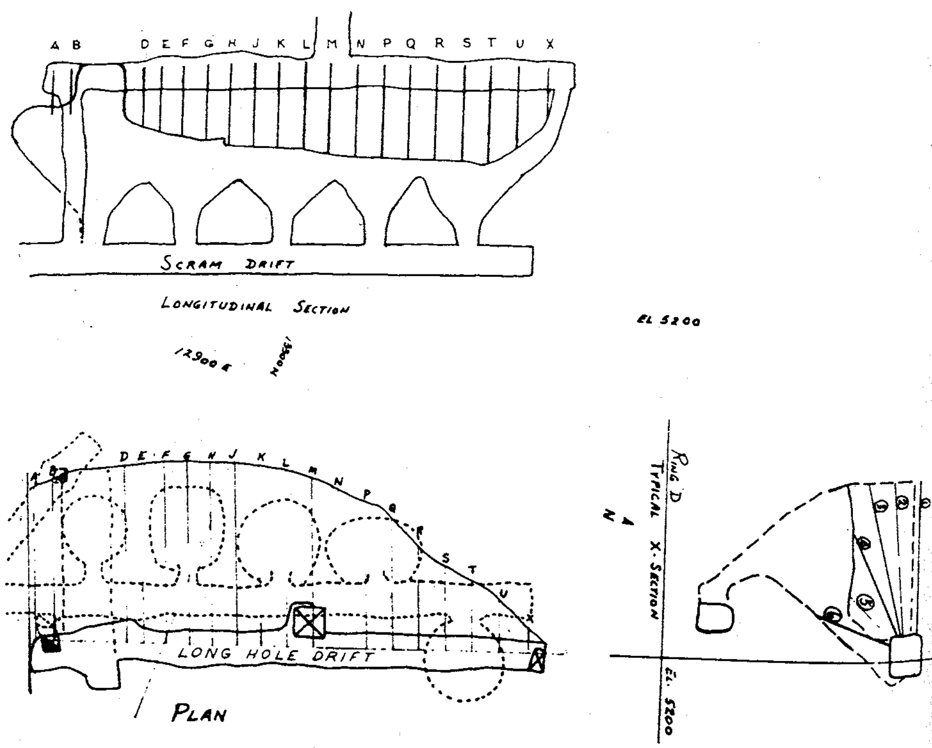




Figure 8. —(above) —The "B" North Stope Undercut.

Figure 7. —(left) —A Cross Section Through the Pillar.

An 8- by 8-foot vertical raise was driven from the undercut to the 5300 level. The raise was drilled off with horizontal rings and four rings were blasted. The Alimak raise platform was then used to extend the raise to the 5520 elevation.

The north stope will be mined to the 5,500-foot level, leaving an 80-foot pillar between the stope back and the bed of the Jordan river.

*South Stope*

The south stope is much shorter than the north and required only two draw points. After coning, the draw points were longholed out to the pillar line and to the end of the ore. An 8- by 8-foot longhole raise was driven from the undercut to the 5400 level. Above the 5400, the orebody splits into two lenses that will be mined by shrinkage methods.

### *Longholing*

Longhole drilling is under contract to Canadian Rock Company Ltd. G.D.C.F. 99 drills are used for drilling the 2⅛-in.-diameter holes. Atlas ⅞-in. hexagonal steel, with 1¼-in. type D thread, is used in 4-ft. lengths. The average length of hole is 60 feet, and rings are drilled with a 6-ft. burden and 9-ft. toe spacing. The rings drilled from the raise are stopped 15 feet short of the pillar line. They are blasted to a 12-ft. sill pillar at the levels. The remaining two rings, as well as the pillar and stope trim-up holes, are then blasted in one shot.

All holes are blown clean before charging with explosive. Five sticks of 1½- by 8-inch, 75 per cent Cilgel are loaded in the toe of all horizontal ring holes. The hole is charged with Amex II. At approximately 20-foot intervals, a stick of 75 per cent Cilgel is placed in the Amex as a booster. All holes are collar primed with S.P. caps and a stick of Cilgel.

Performance to date has been 117 feet drilled per machine shift and 2.9 tons broken per foot drilled, with a powder factor of 0.49 lbs. of combined explosives.

### *Scram — Slushing*

Two I-R 125-hp. electric slushers, back to back, pulling 72-in. Eimco folding scrapers, are used in scrams. Muck to the coarse ore bin passes a 24- by 30-inch grizzly. Amex in plastic bags is used for draw-point bulldozing. One man is normally used for scram operation. Nearly all production to date has been from the north scram, where an average of over 400 tons per shift is maintained.

The total cost of muck delivered to the coarse ore bin is 86 cents per ton. Direct and indirect charges for drilling, loading and blasting total 54 cents per ton. Similarly, scram slushing totals 32 cents per ton.

## Power and Service

Electric power is supplied from the nearby Jordan River power plant of the B.C. Hydro & Power Authority.

A 1,600-foot transmission line was erected to link their forebay to the mine service raise. An armoured cable in the service raise feeds 12,400 volts at 100 amps to the underground transformer station. The transformer station, rock-walled on two sides, with concrete blocks on the two other sides, houses twelve wet-type transformers.

The total connected load is 2,603 hp. The grinding circuit is supplied at 2,300 volts. The remainder of the electric motors, with the exception of fractional horsepowers, are supplied at 440 volts.

Compressed air is supplied by an I-R X.V.H., 1,100-c.f.m. compressor and an I-R P.R.E., 1,100-c.f.m. compressor, both housed in the B.C. Hydro & Power Authority forebay. An 8-in. air line, roughly paralleling the power line, enters the mine through the service raise. Pressure is maintained at 90 p.s.i. throughout the mine.

Water is siphoned from the forebay of the B.C. Power & Hydro Authority. An 8-inch line laid beside the air line supplies water to the mine and mill. Suitable operating pressures are maintained by pressure-reducing valves on the mine and mill supply lines.

## Acknowledgments

The writer wishes to thank the management of Cowichan Copper Co. Ltd. for permission to publish this paper. The assistance of the mine staff and engineering office in the preparation of the paper and drawings is also gratefully acknowledged.